

BPMN (Business Process Model and Notation)

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Overview of BPMN

<https://www.microsoft.com/en-au/microsoft-365/visio/business-process-modeling-notation>

Summary

Business Process Model and Notation (BPMN) is a **standard visual method for representing business processes from start to finish**. It uses standard symbols and modelling techniques so both **technical and non-technical stakeholders** can easily understand and communicate processes.

Main Purpose

- Clearly document and understand business processes
- Improve communication between teams and stakeholders
- Identify inefficiencies and areas for improvement
- Support business process management

Key Benefits

1. **Understanding** – Makes complex workflows easier to understand.
2. **Standardisation** – Provides a common notation that everyone can interpret.
3. **Simplicity** – Uses visual diagrams instead of complicated written descriptions.
4. **Communication** – Bridges communication gaps between teams and departments.
5. **Collaboration** – Allows stakeholders to review and improve processes together.
6. **Better Decisions** – Helps employees understand processes and make decisions faster.

How BPMN Works

BPMN works by using standard symbols and diagrams to visually represent a business process from start to finish. Each symbol has a specific meaning, making the process easy to understand and consistent across different teams.

Key Points

- **Visual flow:** Shows the actions and steps from the beginning to the end of a process.
- **Standardised notation:** Uses globally recognised symbols so technical and non-technical people can understand the process.
- **Bridges communication:** Helps business, technical, internal, and external teams communicate using the same process language.
- **Categorised symbols:** BPMN groups symbols into categories, making diagrams easier to read and interpret.
- **Detailed but simple:** Additional information can be included without making the diagram difficult to understand.
- **Consistent:** BPMN is maintained as a global standard by the **Object Management Group (OMG)**.
- **Diagramming tools:** BPMN diagrams can be created using tools such as Visio, Draw.io, or Lucidchart.

BPMN design elements

BPMN is a collection of symbols and rules used on how to connect those symbols to a business process model. The BPMN language used in those models is based on flowcharts and graphical notations divided into the following five elements.



Flowchart

Flowcharts connect elements that help form a process.

- **Events:** Depicted as circles, these represent an occurrence. Events are broken into three types: Start signaling the first step of the process, intermediate describing any event that occurs between the first and the last, and end signaling the final stage of your business process.
- **Activities:** Visually presented as a rounded corner rectangle, these represent a task performed by a person or system. There are four BPMN activity types: task, sub-process, call, and transaction.
- **Gateways:** Diamond-shaped icons signal if an activity needs to go through a different process flow. A few different gateway symbols separate and connect flows, including exclusive, inclusive, parallel, and complex.



Connecting object

Connecting objects symbolize how objects are connected within the flow.

- **Sequence flows:** Displaying the order of business activities with a solid line and arrow.
- **Message flows:** Messages that "flow" across different departments or organizations, represented by a dashed line with a circle and an arrow.
- **Associations:** Dotted lines connecting artifacts or text to an event, activity, or gateway.
- **Data associations:** These dotted lines with an arrow represent data with flow objects. Data associations show the inputs and outputs of activities.



Swimlane

Swimlanes are graphical icons representing the process participants and who is accountable for which parts.

- **Pools:** These are significant participants, whether within the company or externally, working in the process.
- **Lanes:** These objects show the activities and flow for a participant, dictating their responsibilities within the process. Many lanes can make up a single pool, as a pool is determined by which actions must be completed by that group in the process.



Artifact

An artifact clarifies information added to the diagram while not affecting the flow. Artifacts are associated with a process or connection by an association connection.

- **Text annotation:** Copy providing context to a part of the diagram.
- **Groups:** The logical grouping of activities without changing the flow of the diagram.



Data

Data objects provide the information needed to produce or execute a business process, represented as a file icon. Data objects are also considered artifacts since they don't directly affect the message or sequence flow.

- **Data objects:** Showing what data is necessary for an activity.
- **Data input:** External data input for the entire process.
- **Data output:** The results of that data throughout the entire process.
- **Data stores:** A place to read or write data, lasting throughout the process.

The latest iteration, BPMN 2.0, builds off the previous versions, providing a more extensive set of icons, symbols, and notations, allowing for more detail when necessary.

1. Flowchart

A **flowchart** shows the steps of a business process from start to finish. It uses three main elements:

- **Events (circles):** Show something that happens — **Start, Intermediate, or End.**
- **Activities (rounded rectangles):** Show work or tasks performed by a person or system.
- **Gateways (diamonds):** Show decisions or where the process can split or join, such as **Yes/No** decisions.

Example:

Start → Receive Order → Check Payment → Payment OK? → Process Order → End

2. Connecting Objects

Connecting objects show **how different elements in the process are related or how information flows between them.**

- **Sequence Flow:** Solid line with an arrow showing the order of activities.
- **Message Flow:** Dashed line showing communication between different participants, departments, or organisations.
- **Association:** Dotted line connecting additional information, such as notes or documents, to a process element.
- **Data Association:** Shows how data is provided to or produced by an activity.

Simple way to remember:

Sequence = process flow

Message = communication

Association = additional information

Data association = data input/output

3. Swimlanes

Swimlanes show who is responsible for each part of the process.

- **Pool:** Represents a major participant, such as a **company, customer, or external organisation.**
- **Lane:** Divides a pool into different participants, teams, or roles and shows their responsibilities.

Example:

Customer	Store	Warehouse
Place Order	Confirm Order	Prepare Product
Receive Notification	Prepare Collection	Send to Store

Simple way to remember:

Swimlanes answer: "Who is responsible for this activity?"

4. Artifacts

Artifacts provide additional information or context about the process but do not change the process flow.

Two common examples are:

- **Text Annotation:** Adds a note or explanation to an activity.
- **Group:** Groups related activities together without affecting the process.

Simple way to remember:

Artifacts = extra information that helps explain the process.

5. Data

Data elements show the information that is used, produced, or stored during a business process.

- **Data Object:** Information required or produced by an activity, such as an order form or invoice.
- **Data Input:** Information provided from outside the process.
- **Data Output:** Information produced by the process.
- **Data Store:** A location where information is stored and can be read or updated, such as a database.

Example for Click & Collect:

Customer Order → Process Order → Order Confirmation → Data Store

Here, **Customer Order** is data input, **Order Confirmation** is data output, and the **customer/order database** can be a data store.

Easy way to remember all 5

BPMN Component	Main Question It Answers
Flowchart	What happens?
Connecting Objects	How does it flow/connect?
Swimlane	Who does it?

Swimlane	WHO DOES IT:
Artifact	What additional information explains it?
Data	What information is used or produced?

process maps are generally the most effective form of process documentation because they provide a clear visual representation of workflows, improve communication, and are easier to update and maintain than text-based documentation.

Summary: The Business Ecosystem

Every business process operates within a larger **business ecosystem**. To achieve success, organizations must keep three key elements aligned:

1. **Business Goals** – The objectives the organization wants to achieve (e.g., the CEO's strategic goals).
2. **Processes** – The workflows and activities used to accomplish those goals.
3. **Technology** – The tools and systems that support and improve processes.

People are also a critical part of the ecosystem, but they are considered an underlying component that interacts with all three elements.

Importance of Alignment

Problems occur when one element is overlooked:

- Focusing on **processes and technology** without considering business goals leads to poor outcomes.
- Concentrating on **goals and technology** without well-defined processes reduces effectiveness.
- Having strong **goals and processes** but selecting technology without proper analysis can limit success.

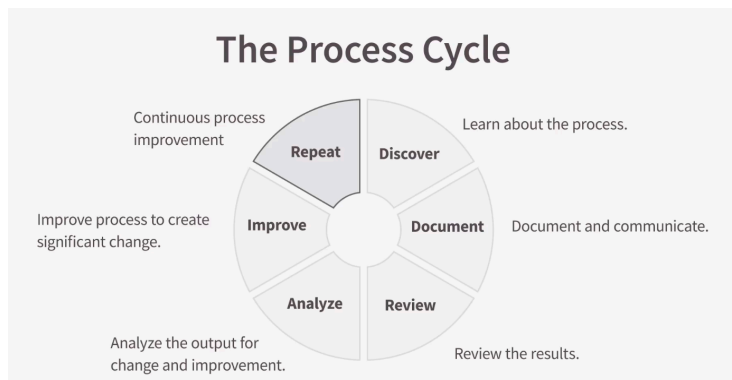
The best results occur when **goals, processes, technology, and people work together in alignment**.

Sonoran Desert Analogy

The instructor compares a business ecosystem to the **Sonoran Desert**, where plants and animals depend on one another to survive. Just as a change in one part of the desert ecosystem can affect the entire system, a weakness or misalignment in business goals, processes, technology, or people can disrupt organizational performance.

Key Takeaway

A successful organization maintains harmony between its **business goals, processes, technology, and people**. Process mapping helps ensure these elements remain aligned, enabling the organization to operate effectively and adapt to change.



Summary: The Business Process Cycle

The **business process cycle** consists of **six phases** that help organizations document, analyze, and improve their processes.

1. **Discover** – Learn about the process by gathering information from process participants and stakeholders to understand how the process actually works.
2. **Document** – Convert the knowledge gathered into documentation. In this course, the focus is on creating **process maps**.
3. **Review** – Validate the process map with participants and stakeholders to ensure it is complete, accurate, and useful.
4. **Analyze** – Examine the process to identify problems such as **bottlenecks, redundant tasks, and inefficient resource usage**.
5. **Improve** – Implement changes and improvements based on the analysis.
6. **Repeat** – Processes continuously change, so the cycle must be repeated to update documentation and make further improvements.

Process Capture

The course focuses mainly on the **Document, Review, Analyze, and Improve** phases, which the instructor refers to as **Process Capture**.

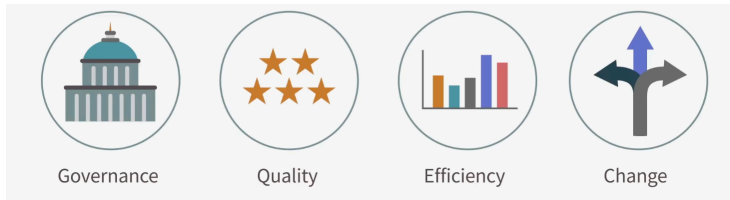
Key Takeaway

The process cycle is **continuous, not a one-time activity**. Organizations should regularly revisit their processes, update their process maps, analyze performance, and implement improvements.

Simple flow:

Discover → Document → Review → Analyze → Improve → Repeat ↺

Motivations for Documenting Processes



Summary: Requirements for a Successful Mapping Project

A successful process mapping project requires proper **support, people, goals, scope, detail, and communication**.

1. Executive Support

- Get **top-level management buy-in**.
- Leadership support is needed for:
 - Resource allocation
 - Budgeting
 - Stakeholder cooperation
 - Supporting and acting on the project results

2. Build the Right Team

A core team should typically have **4–8 people**, including:

- Staff and managers involved in the current process
- **IT representative** to address technology-related aspects
- **Senior leader** who can make decisions and represent management
- **Subject Matter Experts (SMEs)** for areas such as finance, safety, or regulations

3. Define Project Goals

The main goal of the **Discover phase** is to understand the process well enough to create an accurate process map.

Additional goals may include:

- Improving quality
- Increasing efficiency
- Reducing resource usage
- Preparing for process analysis and improvement
- Supporting **knowledge transfer and employee training**

4. Define the Scope

Clearly decide what will be mapped:

- The entire organization's processes
- A process spanning multiple departments
- All processes within one department
- A single specific process

5. Determine the Level of Detail

Decide how much detail each map should contain. A project may use:

- One level of detail
- Two levels
- Three or more levels for complex processes

6. Establish a Communication Plan

Keep both the project team and wider organization informed. This is important because employees may worry that process mapping could **change or redefine their jobs**.

Key Takeaway

Before starting process discovery, make sure you have:

Executive Support → Right Team → Clear Goals → Defined Scope → Appropriate Detail → Communication Plan

These foundations help ensure the process map is accurate, useful, and accepted by the organization.